SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I M. Balaji (198582059) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Balaji

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

cept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept dentification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks:

1. cloud computing fundamentals.

cloud computing

Delivery of computing services over the internet

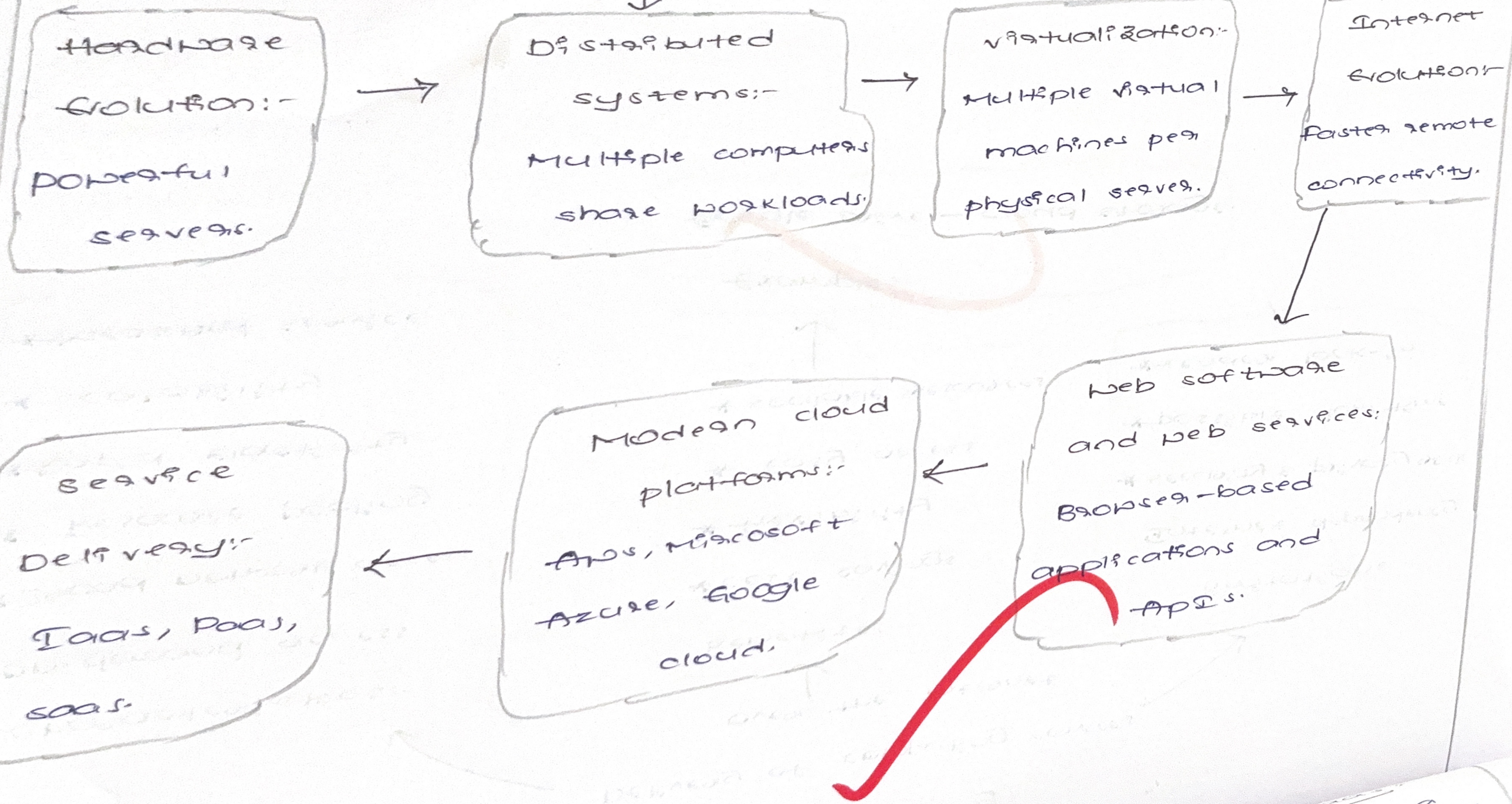
characteristics:-
- on-demand access
- Broad network access
- Resource pooling
- Rapid elasticity
- scalability
- Measured service

Benefits:-
- cost savings
- flexibility
- easy access
- scalable resources

Example:-
Google Drive – cloud storage

limitations:-
- Internet dependency
- security & privacy risk
- possible downtime
- vendor lock-in

2. Evolution of cloud computing.



SaaS:- Application byte application &

Anything as a service various IT services (storage, security, software etc). usage

connectivity.

Web software and web services: Browser-based applications and APIs.

Modern cloud platforms:- AWS, Microsoft Azure, Google cloud.

Service Delivery:- Iaas, Paas, saas.

3. cloud service models.

Iaas:-

Infrastructure layer provider manages:-
* Hardware, network virtualization

User manages: OS, applications, data

Paas:-

platform layer provider manages:
* Infrastructure, OS, runtime/platform

User manages:
* Application, data.

saas:-

Application layer
* Application & underlying infrastructure
* uses software & manages content/ settings

xaas:-

Anything as a service
* various IT services (storage, security, software etc).
* service usage and configuration.

provides (infrastructure & platform).

shared Responsibilitys depends on service model.

consumes application and data.

Q. virtualisation and webservices as enablers of cloud

virtualisation + web services.

virtualisation.

physical servers
↓ creates/manage
hypervisor
↓ creates
virtual machines
↓ abstracts
Abstraction and efficient resource use

web services

API
↓ enables communication
SOAP and REST
↓
Application-to-service communication

→ cloud deployment and scalability

- Deploy applications and resources
- Increase/decrease resources as demand changes
- provider supplies service
- consumer access/uses service.

13/10/26